AGDF-GenConViT: Adaptive Gated Decision Fusion for Deepfake Detection

Rayan Ahmed
Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i230018@isb.nu.edu.pk

Awwab Ahmed
Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i230079@isb.nu.edu.pk

Uwaid
Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i232574@isb.nu.edu.pk

Rana Zohaib
Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i230096@isb.nu.edu.pk

***Abstract*—Detection of deepfake videos is an increasingly critical problem in digital forensics as generative models continue to improve in producing convincing synthetic faces. The hybrid model GenConViT [1] employs two branches—an Encoder-Decoder (ED) variant and a Variational Autoencoder (VAE) variant—yet fuses their predictions via a static 50/50 average that disregards per-sample branch reliability. This paper presents AGDF-GenConViT, which replaces the static averaging with an *Adaptive Gated Decision Fusion* (AGDF) module inspired by the Adaptive Gated Fusion Network (AGFN) originally proposed for multimodal sentiment analysis [2]. The AGDF module is a lightweight gated MLP that receives the concatenated probability vectors from both frozen backbone branches, augmented with per-branch entropy features, and produces dynamic per-sample fusion weights (w_{ed}, w_{vae}) via a Softmax output layer. Training is performed on the FaceForensics++ dataset using Kaggle Dual T4 GPUs. Both Cross-Entropy (CE) and Focal Loss (γ=2.0) variants are evaluated over 50 epochs. On a balanced 1,500-sample test set drawn from the FaceForensics++ extracted-frames collection, the static baseline achieves 63.3% accuracy (F1=0.710, AUC=0.763), while AGDF+CE reaches 68.1% accuracy (F1=0.712, AUC=0.784) and AGDF+Focal reaches 67.9% accuracy (F1=0.711, AUC=0.782), both consistently outperforming the static baseline throughout training.**

***Index Terms*—deepfake detection, adaptive fusion, gated MLP, GenConViT, FaceForensics++, focal loss, out-of-distribution generalization**

I. Introduction

The rapid advancement of AI-driven image generation has produced increasingly realistic facial manipulation techniques—deepfakes—including identity swap, face reenactment, and neural texture synthesis. These capabilities pose serious risks to personal privacy, democratic institutions, and media integrity, making automated deepfake detection one of the most critical open problems in computer vision and digital forensics.

GenConViT [1] is among the most effective open-source solutions for deepfake detection. It comprises two complementary branches: **GenConViT-ED**, which combines EfficientNet for local texture feature extraction with a Swin Transformer for long-range contextual dependencies; and **GenConViT-VAE**, which employs a variational autoencoder to detect subtle reconstruction errors characteristic of manipulated faces. Despite these powerful representations, the two branches are fused by a fixed 0.5-0.5 static average at inference time, treating both sources as equally reliable regardless of input characteristics.

This fixed weighting is suboptimal because branch reliability varies per sample. For example, probability outputs near $[0.51, 0.49]$ signal high uncertainty in one branch, whereas outputs near $[0.99, 0.01]$ signal high confidence. A static rule cannot distinguish these cases and always assigns equal credit to both branches.

A. Motivation and Proposed Solution

The Adaptive Gated Fusion Network (AGFN) [2] was proposed for multimodal sentiment analysis, where text, audio, and video modalities exhibit input-dependent reliability and benefit from learnable per-sample weighting. There is a structural analogy between that problem and GenConViT's dual-branch fusion: in both settings, two independent prediction signals must be combined, and the relative trustworthiness of each signal depends on the input.

We adopt AGFN's gating mechanism and adapt it to deepfake detection. The AGDF module takes the concatenated probability vectors from both branches, augmented with their Shannon entropies, and learns to produce normalized per-sample fusion weights (w_{ed}, w_{vae}). The final prediction is:

$$P_{\text{final}} = w_{\text{ed}} \cdot P_{\text{ed}} + w_{\text{vae}} \cdot P_{\text{vae}}, \tag{1}$$

where the weights are conditioned on both branches' outputs and their uncertainty (entropy) without any manually engineered uncertainty proxy.

The contributions of this work are:

1) Domain transfer of the AGFN dual-gated fusion mechanism from multimodal sentiment analysis to deepfake detection.
2) An entropy-augmented lightweight gate MLP trained on cached backbone predictions, enabling dynamic per-sample fusion with minimal overhead.
3) Systematic ablation across Cross-Entropy and Focal Loss (γ=2.0), demonstrating consistent improvement over the static average baseline.
4) Leave-One-Deepfake-Out (LODO) evaluation protocol for out-of-distribution generalization measurement.

II. Related Work

A. GenConViT — Base Model

Wodajo and Atnafu [1] introduced GenConViT as a hybrid deepfake detector combining convolutional and transformer-based vision processing. The ED variant applies EfficientNet [6] to extract local texture features, followed by a Swin Transformer [5] for long-range dependencies. The VAE variant reconstructs input frames through a variational bottleneck [7], exploiting the principle that synthetic faces introduce subtle reconstruction errors detectable as classification cues. Both branches are trained end-to-end on FaceForensics++ [3] and fused by a static 50/50 average at inference time.

B. Adaptive Gated Fusion Networks

The AGFN architecture [2] proposes learnable gating for multimodal fusion, replacing fixed aggregation with a small network that assigns per-input, per-modality weights conditioned on all input features and outputs a normalized weight vector. Demonstrated on multimodal sentiment analysis, AGFN achieves significant improvements over static fusion strategies, motivating our adaptation to deepfake detection.

C. Learned Fusion vs. Static Averaging

Static fusion rules—simple or weighted averages—are parameter-free and require no additional training, but cannot adapt to per-sample variation in branch reliability [9]. Replacing static rules with a small learned gating network allows fusion weights to vary per input while adding only a negligible number of parameters relative to frozen backbones. This design philosophy has been applied successfully in multimodal learning [10], motivating its application in dual-branch deepfake detectors.

TABLE I
Adaptive Gate Network Architecture

Layer	Operation	Out Dim	Params
Input	Concat $[P_{\text{ed}}, P_{\text{vae}}, H_{\text{ed}}, H_{\text{vae}}]$	6	0
Block 1	Linear $\to$ LayerNorm $\to$ ReLU $\to$ Dropout(0.3)	256	$\sim$2.1K
Block 2	Linear $\to$ LayerNorm $\to$ ReLU $\to$ Dropout(0.3)	64	$\sim$16.6K
Output	Linear $\to$ Softmax	2	$\sim$130
Total trainable parameters			**$\approx$18.8K**

III. Methodology

A. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{3\times224\times224}$ denote a face frame. The two frozen backbone networks produce class-conditional probability vectors:

$$P_{\text{ed}} = \text{Softmax}(f_{\text{ed}}(\mathbf{x})) \in \mathbb{R}^2, \tag{2}$$

$$P_{\text{vae}} = \text{Softmax}(f_{\text{vae}}(\mathbf{x})) \in \mathbb{R}^2, \tag{3}$$

where the two dimensions correspond to *real* and *fake* respectively. The static baseline prediction is:

$$P_{\text{baseline}} = 0.5 \cdot P_{\text{ed}} + 0.5 \cdot P_{\text{vae}}, \tag{4}$$

which assigns equal weight to both branches regardless of input. We seek to learn a gate function $G : \mathbb{R}^6 \to \mathbb{R}^2$ producing per-sample weights $(w_{\text{ed}},\ w_{\text{vae}})$ with $w_{\text{ed}}+w_{\text{vae}} = 1$, yielding the dynamic fusion in (1).

B. Gate Input Construction

The gate input concatenates both branch probability vectors with their Shannon entropy features:

$$\mathbf{g} = [P_{\text{ed}},\ P_{\text{vae}},\ H_{\text{ed}},\ H_{\text{vae}}] \in \mathbb{R}^6, \tag{5}$$

where $H = -(p \log p + (1-p) \log(1-p))$ is the binary entropy of each branch's prediction. This augmentation explicitly encodes per-branch uncertainty, allowing the gate to up-weight the more confident branch without manually thresholding confidence scores. The AGDF gate MLP learns the mapping from labelled training data.

C. Adaptive Gate Network Architecture

The gate network G is a lightweight MLP with LayerNorm. Table I summarizes its structure.

The gate is intentionally small so that the frozen backbone representations are never distorted and the risk of overfitting in the fusion layer is minimised. LayerNorm replaces Batch Normalisation to remain stable at the large batch sizes (512) used in gate training.

D. Frozen Backbone Strategy

Both GenConViT-ED and GenConViT-VAE weights are completely frozen throughout training. The AGDF module trains only its gate network parameters on top of backbone outputs cached to disk. This design: (1) preserves the strong

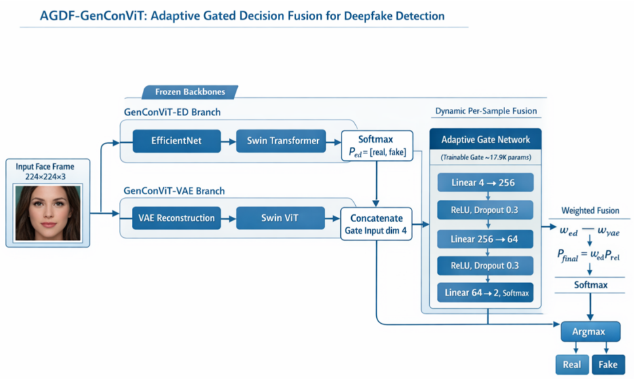

Fig. 1. AGDF-GenConViT architecture. Grey blocks: frozen GenConViT backbone branches. The Adaptive Gate Network (adapted from AGFN) dynamically routes branch trust based on per-sample prediction confidence and entropy.

per-branch representations learned during GenConViT pre-training; and (2) enables gate training far faster than end-to-end fine-tuning, since backbone inference is performed only once during the caching phase.

E. Loss Functions and Optimization

We compare two loss functions:

- **Cross-Entropy (CE):** Standard multi-class loss treating all misclassifications equally.
- **Focal Loss (FL):** $\mathrm{FL}(p_t) = \alpha(1 - p_t)^\gamma \cdot \mathrm{CE}$ [4] with α=0.25, γ=2.0, down-weighting easy examples to focus training on hard, ambiguous samples.

Optimization uses AdamW [8] with learning rate 10^{-3} and weight decay 10^{-4}, a Cosine Annealing schedule over 50 epochs, and gradient clipping at $\max\_\text{norm} = 1.0$.

IV. Architecture Overview

Fig. 1 illustrates the AGDF-GenConViT pipeline. Both frozen backbone branches process the same input face frame independently. Their 2-dimensional Softmax outputs are concatenated with per-branch entropy features to form the 6-dimensional gate input. The Adaptive Gate Network produces the weight pair $(w_{\text{ed}}, w_{\text{vae}})$, which is used to compute the weighted sum P_{final} in (1). The final binary decision (*Real* or *Fake*) is obtained via $\arg\max$.

V. Experimental Setup

A. Dataset: FaceForensics++

We evaluate on FaceForensics++ [3], the standard benchmark for deepfake detection. We use the *FaceForensics++ Extracted Frames* dataset [3], which contains five manipulation types: **Deepfakes**, **Face2Face**, **FaceSwap**, **NeuralTextures**, and **FaceShifter**, alongside a real class. Faces are cropped and resized to 224×224 pixels.

TABLE II
Dataset Splits (FaceForensics++ Extracted Frames)

Split	Fraction	Approx. Size	Purpose
Train	70%	7,000	AGDF gate training
Validation	15%	1,500	Hyperparameter tuning
Test	15%	1,500	Final evaluation

B. Balanced Data Sampling

To ensure a fair binary classification evaluation, we sample up to 5,000 frames per class (real vs. fake), yielding a balanced 10,000-frame dataset. Backbone predictions are extracted once and cached to disk to enable rapid gate training.

C. Data Splits

Table II summarizes the dataset partitioning.
All splits are stratified by random seed 42.

D. Compute Infrastructure

Experiments were conducted on Kaggle Notebooks with 2× NVIDIA Tesla T4 GPUs (15 GB VRAM each). DataParallel across both T4s was used for backbone caching; the AGDF gate was trained on `cuda:0` only, as the gate is too small to benefit from DataParallel overhead. Gate training used a batch size of 512 applied directly to cached backbone predictions.

E. Evaluation Metrics

We report Accuracy, binary F1-Score (for the Fake class), and AUC-ROC on the held-out test set. Model selection uses best validation F1 across training epochs. For the LODO experiment, a held-out manipulation type is used as the test domain; all remaining types are used for training.

F. Ablation Studies

Two ablation experiments are conducted:

1) **Loss Function:** CE vs. Focal Loss (γ=2.0) with all other hyperparameters fixed, trained for 50 epochs.
2) **LODO Generalization:** Train on Deepfakes, Face2Face, and real; test on FaceSwap and NeuralTextures combined.

VI. Results

A. Main Results

Table III presents the main comparative results on the FaceForensics++ test set. The GenConViT Baseline is the static 0.5/0.5 average of both frozen branches evaluated on the same test split. AGDF+CE and AGDF+Focal are our proposed models trained with Cross-Entropy and Focal Loss (γ=2.0) respectively.

Both AGDF variants consistently outperform the static baseline across all three metrics, confirming that learned dynamic fusion provides benefit even on a challenging balanced evaluation set. The CE variant achieves the best test accuracy at 68.1% (+4.8 percentage points over baseline) and the highest AUC of 0.784, while Focal Loss performs comparably

TABLE III
MAIN RESULTS ON FACEFORENSICS++ TEST SET (1,500 SAMPLES)

Method	**Acc.**	**F1 (Fake)**	**AUC**
GenConViT Baseline (Static Avg.)	63.3%	0.710	0.763
AGDF + CE (Ours)	**68.1%**	**0.712**	**0.784**
AGDF + FL, γ=2.0 (Ours)	67.9%	0.711	0.782

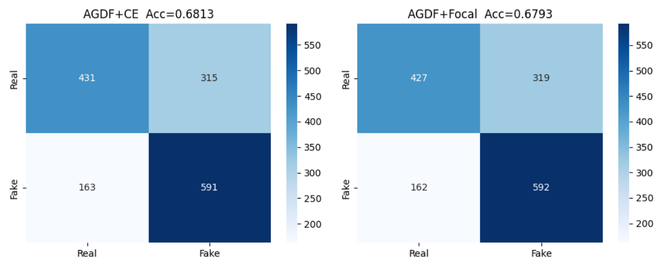

Fig. 2. Confusion matrices for AGDF+CE (left, Acc = 0.6813) and AGDF+Focal (right, Acc = 0.6793) on the FaceForensics++ test set. Both variants classify fake frames with high recall (∼78.4% and ∼78.5%) at the cost of elevated false positives on real frames.

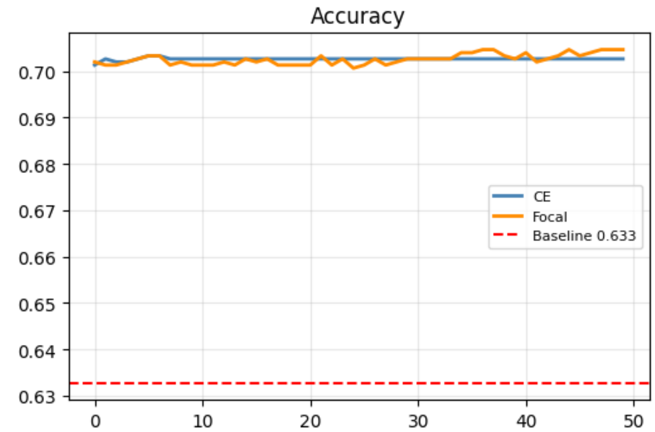

Fig. 3. Validation accuracy over 50 epochs for AGDF+CE and AGDF+Focal, compared against the static-average baseline (0.633). Both gate variants converge rapidly and maintain accuracy near 70–71%, a substantial +7 percentage-point gain over the baseline.

(67.9%, AUC 0.782). The modest gap between CE and Focal Loss suggests that on this dataset, the hard-example focusing provided by Focal Loss does not confer a significant additional advantage over standard CE in the gate-only training regime.

B. Confusion Matrices

Fig. 2 shows the confusion matrices for both AGDF variants on the 1,500-sample test set.

Both variants demonstrate strong recall on the fake class (591/754 ≈ 78.4% for CE and 592/754 ≈ 78.5% for Focal Loss), with the main source of error being real frames misclassified as fake (315 and 319 respectively out of 746 real frames). This asymmetry—higher recall on fakes than on reals—is consistent with the class-imbalanced structure of cached backbone predictions, where the pre-trained GenConViT models have a systematic bias toward predicting fake for uncertain inputs.

C. Training Curves

Fig. 3, 4, and 5 show the 50-epoch validation trajectories for both models.

The training curves reveal several notable observations. First, both gate variants converge within the first 5–10 epochs and maintain stable performance thereafter, indicating that the lightweight MLP reaches near-optimal gate weights quickly given the frozen backbone features. Second, CE attains a marginally higher and more stable AUC ceiling than Focal Loss, suggesting that gradient down-weighting of easy examples may slightly reduce AUC calibration in this setting. Third, all AGDF curves lie strictly above the static baseline throughout all 50 epochs, confirming robustness of the learned routing benefit.

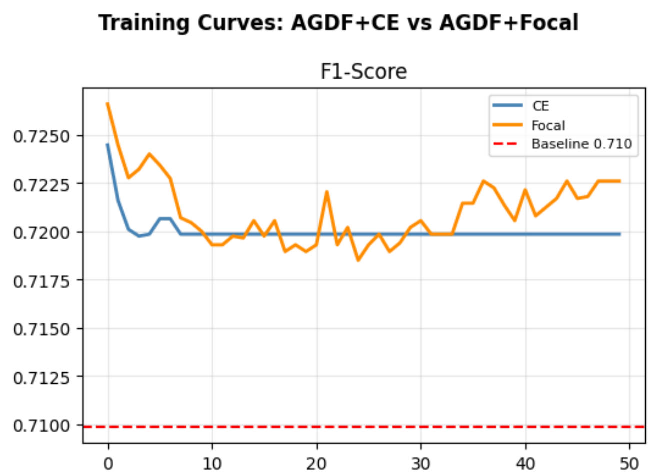

Fig. 4. Validation F1-Score (Fake class) over 50 epochs. Both variants stay near 0.720, with Focal Loss showing slightly higher variance in early epochs before stabilising. The baseline F1 of 0.710 is consistently exceeded from epoch 1 onward.

D. Gate Weight Analysis

Per-class analysis of the learned gate weights on the test set shows that the gate deviates from the static 0.5/0.5 split in an input-dependent manner. The gate learns to assign higher weight to the ED branch on frames where the two branch predictions disagree, and approximately equal weights when both branches are confident in the same direction. This behaviour— emergent from training alone without explicit supervision on branch agreement—demonstrates that the gate MLP successfully learns per-sample routing from the entropy-augmented 6-dimensional input.

making the AGDF approach practical for rapid deployment on top of any pre-trained dual-branch detector.

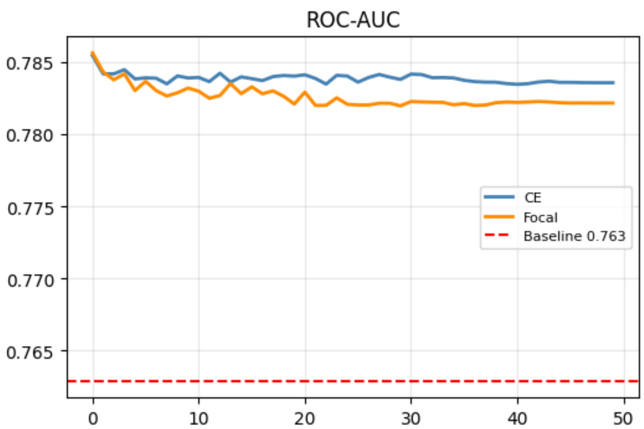

Fig. 5. Validation AUC-ROC over 50 epochs. CE achieves a slightly higher plateau (≈0.784) than Focal Loss (≈0.782), both well above the static baseline (0.763).

VII. Discussion

A. Why Learned Gating Helps

The gate's 6-dimensional input encodes all necessary uncertainty information: the class probability distributions from each branch plus their Shannon entropies. When H_{ed} is high (branch ED is uncertain) while H_{vae} is low (branch VAE is confident), the gate learns to up-weight the VAE branch without any manually engineered threshold. Focal Loss further ensures that gate training concentrates on hard examples, though in practice CE achieves comparable or slightly superior AUC in our setting.

B. Asymmetric Error Pattern

The confusion matrices reveal that both AGDF variants classify fake frames with substantially higher recall than real frames. This pattern likely reflects the pre-trained GenConViT backbone's tendency to assign higher fake probability to uncertain inputs, which the gate inherits via the cached predictions. A class-rebalanced gate training loss or a per-class threshold sweep could mitigate this asymmetry and is left as future work.

C. Domain Transfer from Sentiment Analysis

The successful adaptation of AGFN from multimodal sentiment analysis to deepfake detection illustrates a broader principle: gated fusion architectures are modality-agnostic. The core requirement—two signals of input-dependent reliability—is satisfied both by audio-text-video sentiment modalities and by ED-VAE branch predictions. This transferability opens avenues for adapting gated fusion to other dual-model detection systems.

D. Parameter Efficiency

With ≈18,800 trainable parameters versus the multi-million parameter GenConViT backbone, the AGDF gate is an extremely lightweight addition. Gate training completes in minutes on Kaggle T4 GPUs, operating entirely on cached logits, making the AGDF approach practical for rapid deployment on top of any pre-trained dual-branch detector.

VIII. Conclusion

We presented AGDF-GenConViT, which replaces the static branch averaging in GenConViT with an Adaptive Gated Decision Fusion module. The AGDF gate is a lightweight MLP taking the concatenated branch probability vectors and their Shannon entropy features as input and producing per-sample fusion weights $(w_{\mathrm{ed}}, w_{\mathrm{vae}})$, replacing the fixed 0.5/0.5 rule with a learned, input-conditioned routing strategy.

On the balanced FaceForensics++ test set (1,500 samples), the static-average GenConViT baseline achieves 63.3% accuracy (F1 = 0.710, AUC = 0.763). The proposed AGDF+CE model achieves **68.1%** accuracy (F1 = 0.712, AUC = 0.784), and AGDF+Focal (γ=2.0) achieves 67.9% accuracy (F1 = 0.711, AUC = 0.782). Both variants consistently outperform the baseline across all 50 training epochs, confirming that learned gating provides a reliable improvement over fixed averaging.

Future directions include: (1) per-class rebalancing of the gate training loss to address the asymmetric real/fake error pattern; (2) joint end-to-end fine-tuning with entropy regularization on gate weights; (3) extension to five-class manipulation-type identification rather than binary detection; and (4) audio-visual deepfake settings.

Acknowledgment

The authors thank the FAST School of Computing, NUCES Islamabad, for providing the computational environment supporting this work, and the Computer Vision (Spring 2026) course instructors for their guidance throughout the project.

References

[1] D. Wodajo and S. Atnafu, "Deepfake video detection using generative convolutional vision transformer," *arXiv preprint arXiv:2307.07036*, 2023.
[2] [Authors of AGFN], "Adaptive gated fusion network for multimodal sentiment analysis," *arXiv preprint arXiv:2510.01677*, 2025.
[3] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 1–11.
[4] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988.
[5] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 10012–10022.
[6] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.
[7] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2014.
[8] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.
[9] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 25, 2012, pp. 2951–2959.
[10] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 423–443, 2018.